

Linear Programming Report

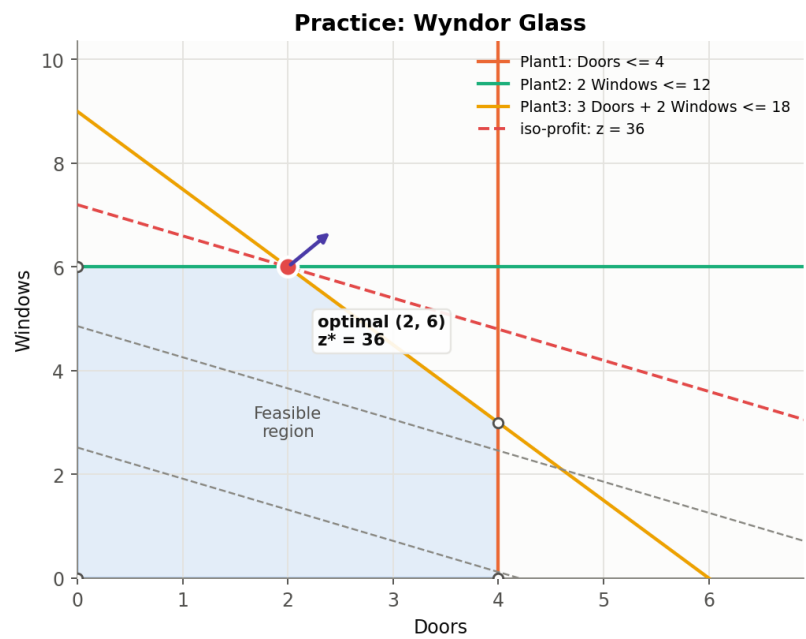
Sensitivity Analysis & Duality

Practice: Wyndor Glass

Two new products (doors x_1 , windows x_2) share capacity in three plants. Profit in \$1000 per batch.

Source: Practice problem (classic textbook example)

Problem type	MAX, 2 variables, 3 constraints
Status	OPTIMAL
Optimal Profit (\$1000)	36
Dual status	OPTIMAL
Prepared by	Dr. Islam Ali
Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	27 September 2026, 15:11



1. Model formulation

```
MAX  z = 3 Doors + 5 Windows
subject to
    Doors  <=  4      (Plant1)
    2 Windows <= 12      (Plant2)
    3 Doors + 2 Windows <= 18  (Plant3)
    Doors >=0, Windows >=0
```

Data table

Constraint	Doors	Windows	Sense	RHS
Plant1	1	0	<=	4
Plant2	0	2	<=	12
Plant3	3	2	<=	18
Objective c	3	5	max	—

Definitions

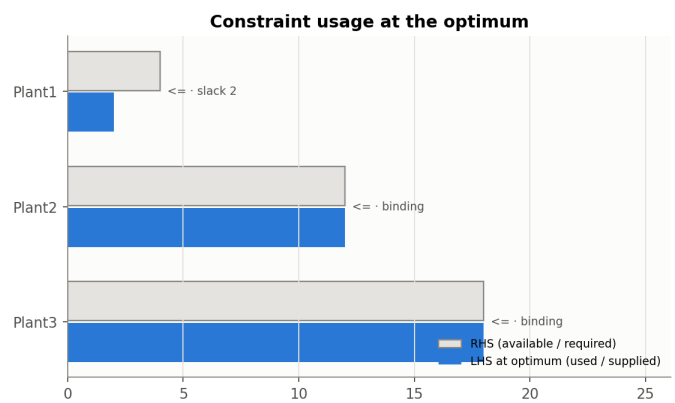
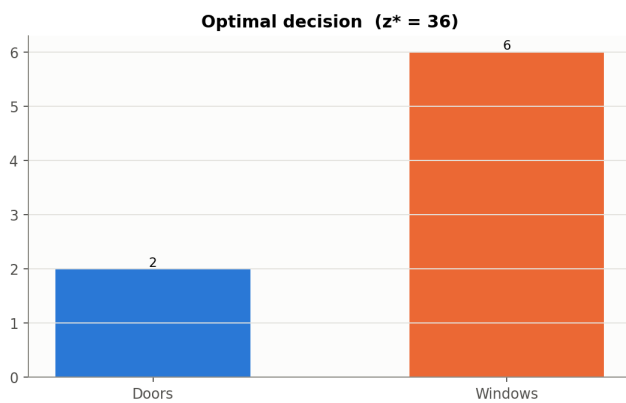
- **Doors**: batches of doors
- **Windows**: batches of windows
- **Plant1**: hours in plant 1
- **Plant2**: hours in plant 2
- **Plant3**: hours in plant 3

2. Optimal solution

Optimal Profit (\$1000): **36**

Variable	Description	Optimal value	Obj. coefficient	Contribution
Doors	batches of doors	2	3	6
Windows	batches of windows	6	5	30

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Plant1	2	$\leq$	4	2	no
Plant2	12	$\leq$	12	0	yes
Plant3	18	$\leq$	18	0	yes



Corner points

Doors	Windows	Profit (\$1000)	Optimal?
0	0	0	
4	0	12	
4	3	27	
2	6	36	✓
0	6	30	

3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
Doors	2	0	3	4.5	3	0	7.5
Windows	6	0	5	∞	3	2	∞

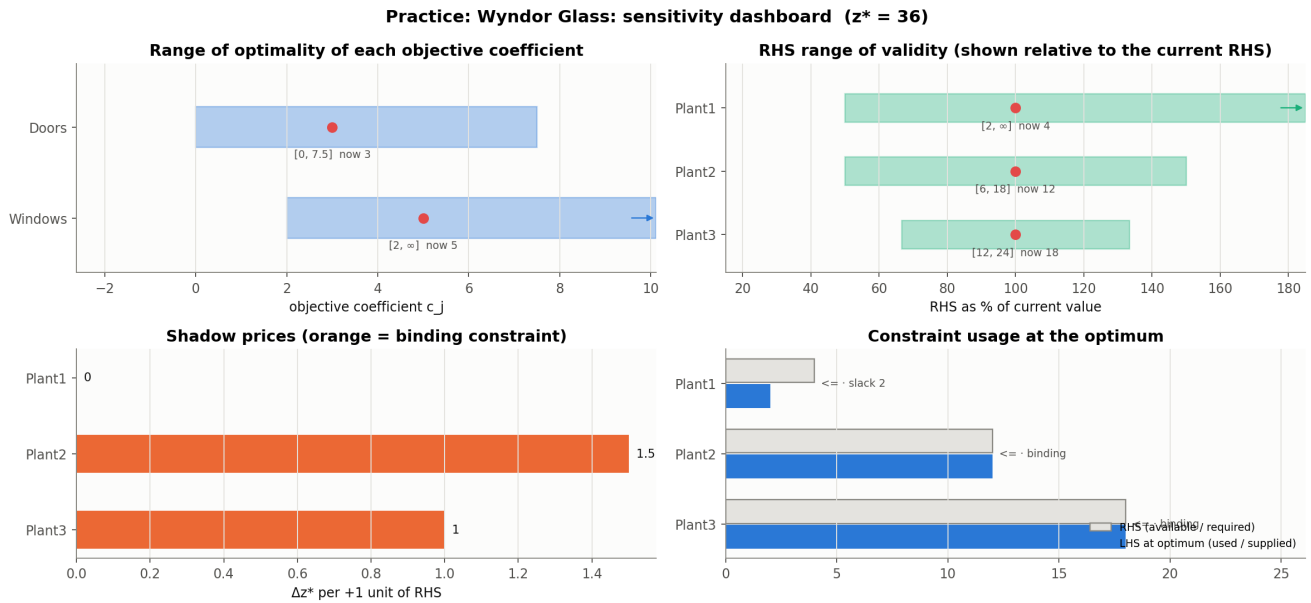
Right-hand-side ranging (constraints)

Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Plant1	2	$\leq$	4	2	0	∞	2	2	∞
Plant2	12	$\leq$	12	0	1.5	6	6	6	18
Plant3	18	$\leq$	18	0	1	6	6	12	24

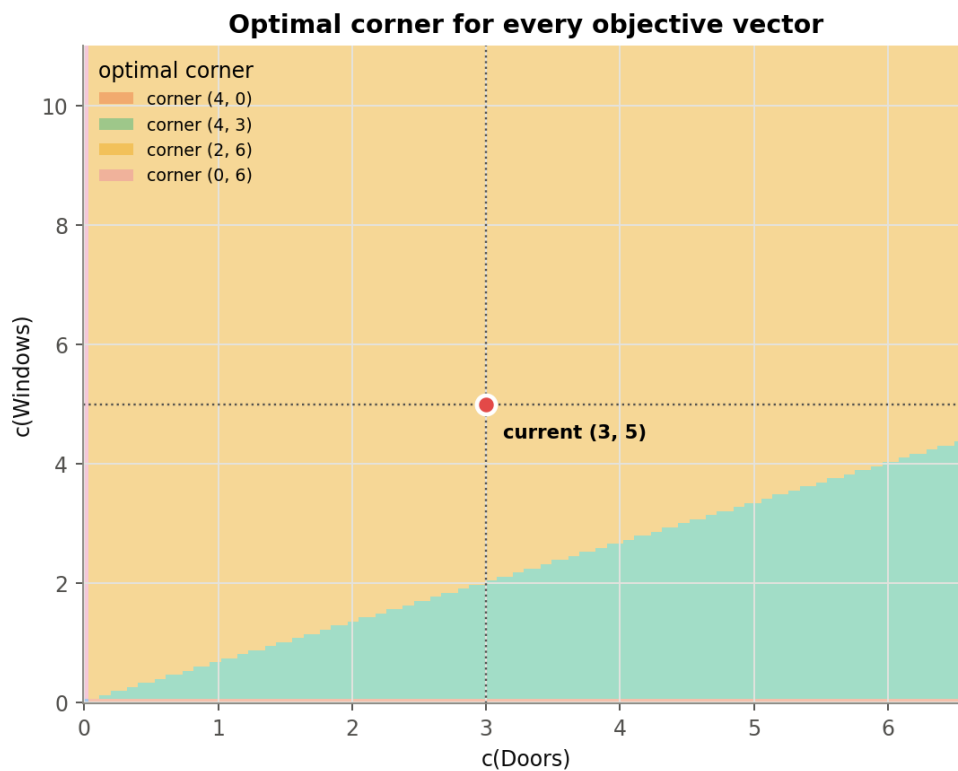
Interpretation

- **Doors** = 2. Its objective coefficient (3) may vary in $[0, 7.5]$ without changing the optimal solution (one change at a time).
- **Windows** = 6. Its objective coefficient (5) may vary in $[2, \infty]$ without changing the optimal solution (one change at a time).
- **Plant1** is not binding (slack 2), so its shadow price is 0; the RHS can move within $[2, \infty]$ without effect on Profit (\$1000).
- **Plant2** is binding. Shadow price 1.5: each extra unit of RHS changes Profit (\$1000) by 1.5, valid while the RHS stays in $[6, 18]$.
- **Plant3** is binding. Shadow price 1: each extra unit of RHS changes Profit (\$1000) by 1, valid while the RHS stays in $[12, 24]$.

Sensitivity dashboard

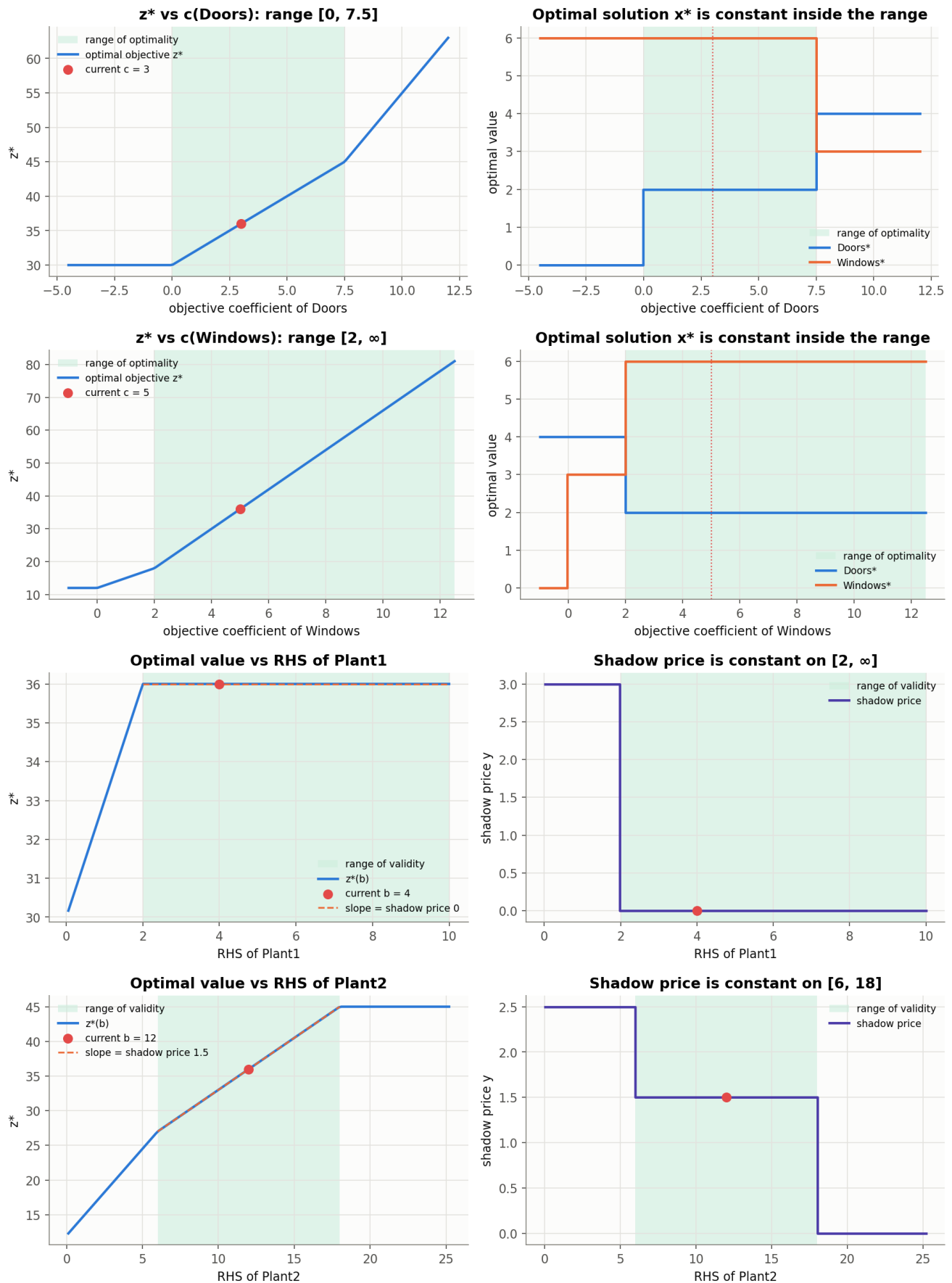


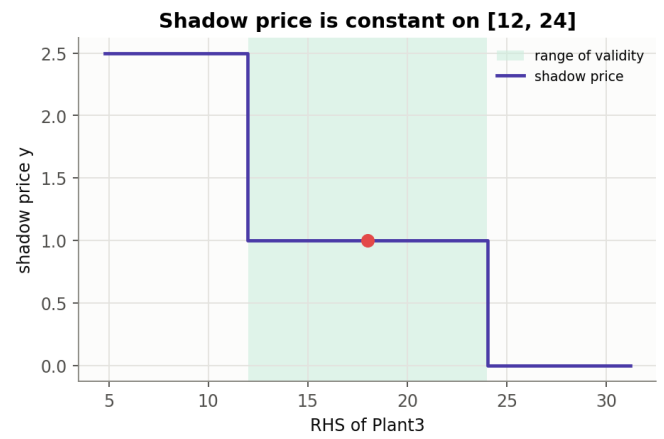
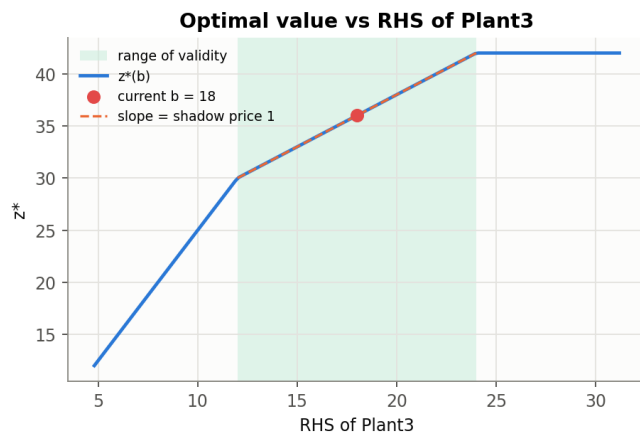
Optimal corner for every objective vector



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.



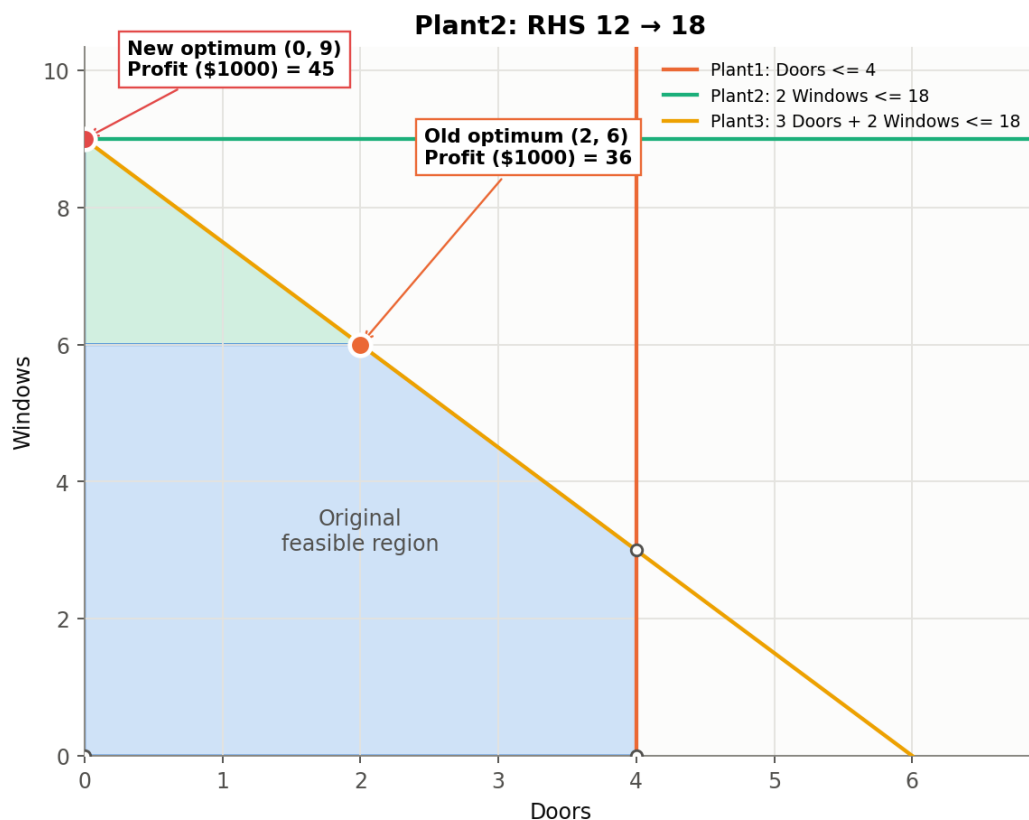
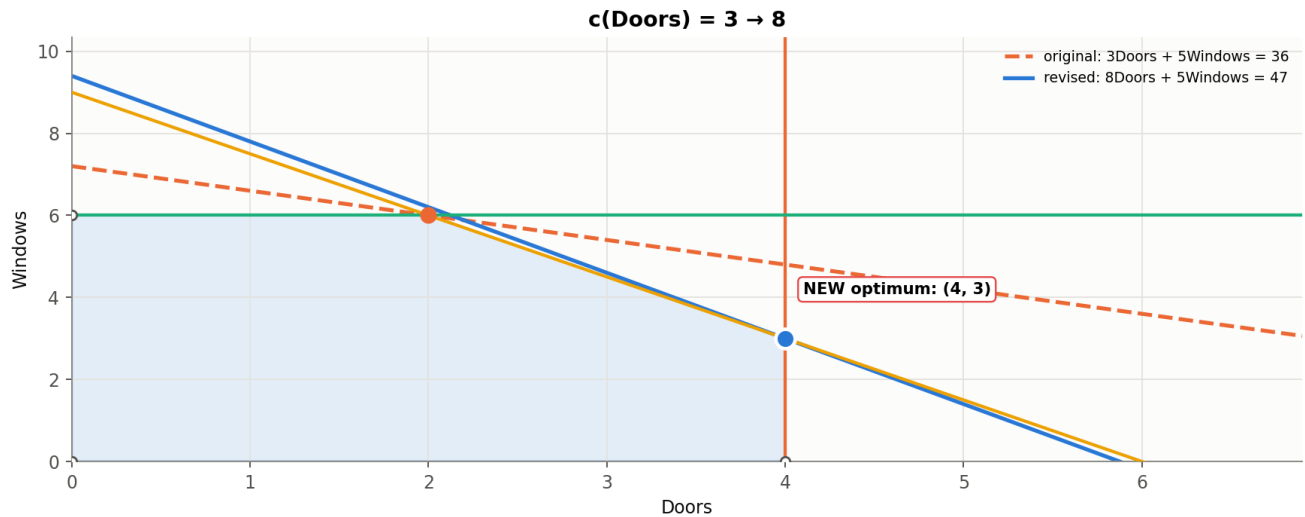


5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(Doors): 3 → 8	no	–	47	4, 3	door profit rises to 8
b(Plant2): 12 → 18	yes	45	45	0, 9	plant 2 capacity rises to 18

Changing an objective coefficient rotates the iso-profit line; the feasible region never moves



6. Duality

Dual formulation

MIN $z = 4 y_{\text{Plant1}} + 12 y_{\text{Plant2}} + 18 y_{\text{Plant3}}$
 subject to
 $y_{\text{Plant1}} + 3 y_{\text{Plant3}} \geq 3$ (Doors)
 $2 y_{\text{Plant2}} + 2 y_{\text{Plant3}} \geq 5$ (Windows)
 $y_{\text{Plant1}} \geq 0, y_{\text{Plant2}} \geq 0, y_{\text{Plant3}} \geq 0$

Dual solution (resource / requirement prices)

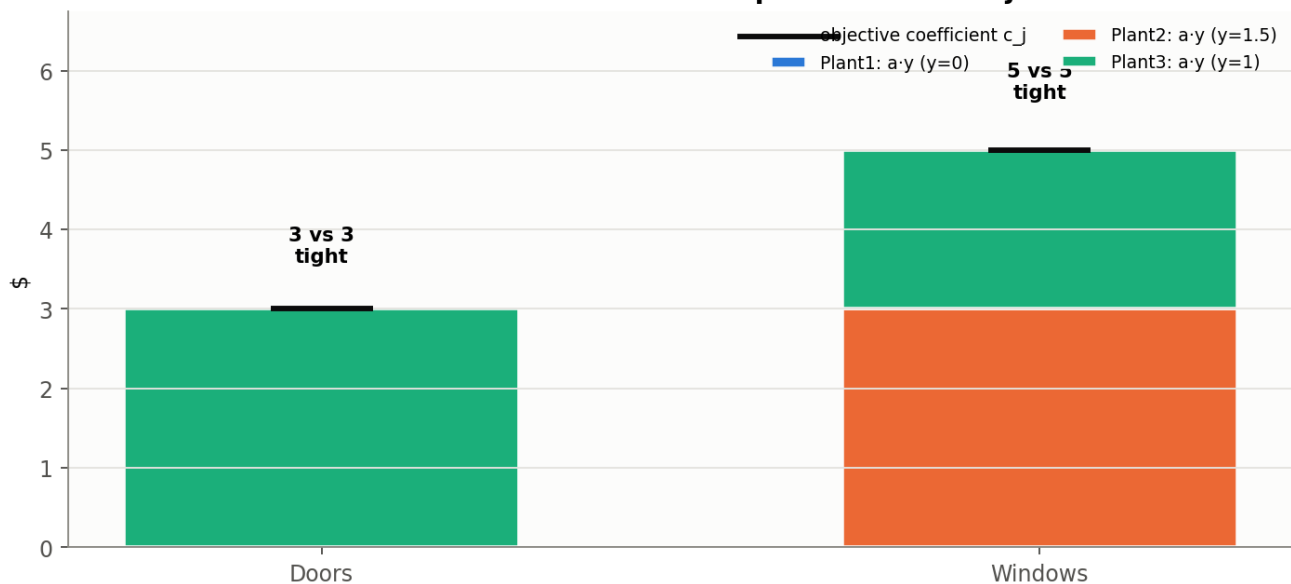
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{Plant1}	Plant1	0	0	0
y_{Plant2}	Plant2	1.5	1.5	18
y_{Plant3}	Plant3	1	1	18

Strong duality check: primal $z^* = 36$, dual $w^* = 36$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Plant1 slack $\times y_{\text{Plant1}}$	2	0	0
Plant2 slack $\times y_{\text{Plant2}}$	0	1.5	0
Plant3 slack $\times y_{\text{Plant3}}$	0	1	0
Doors $\times$ surplus of dual constraint Doors	0	2	0
Windows $\times$ surplus of dual constraint Windows	0	6	0

Dual constraints: value of resources in each product vs its objective coefficient

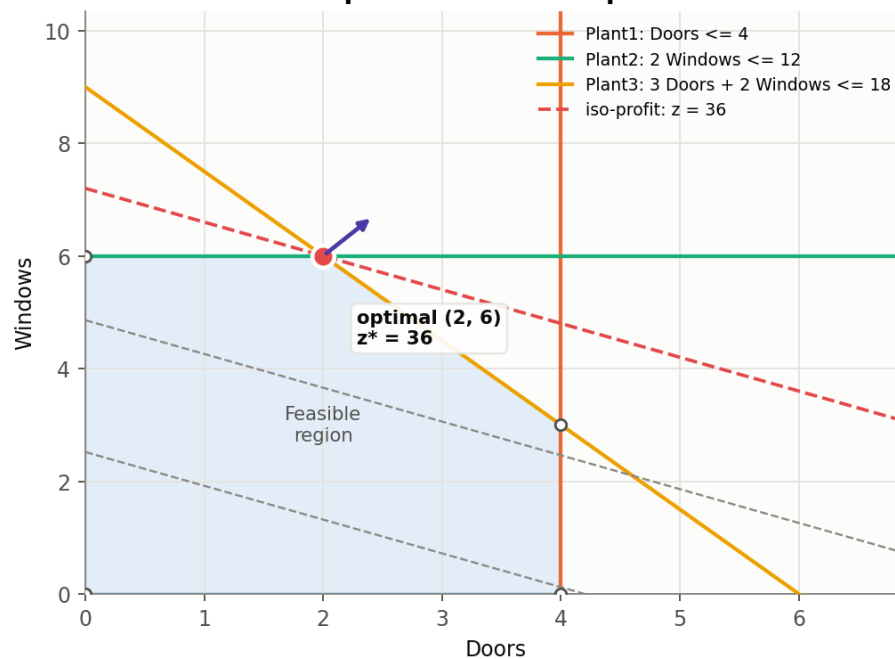


Primal (max): 3 constraints × 2 variables

3	5	← objective
1	0	≤ 4
0	2	≤ 12
3	2	≤ 18
Doors	Windows	

Dual (min): 2 constraints × 3 variables

4	12	18	← objective
1	0	3	≥ 3
0	2	2	≥ 5
y_Plant1	y_Plant2	y_Plant3	

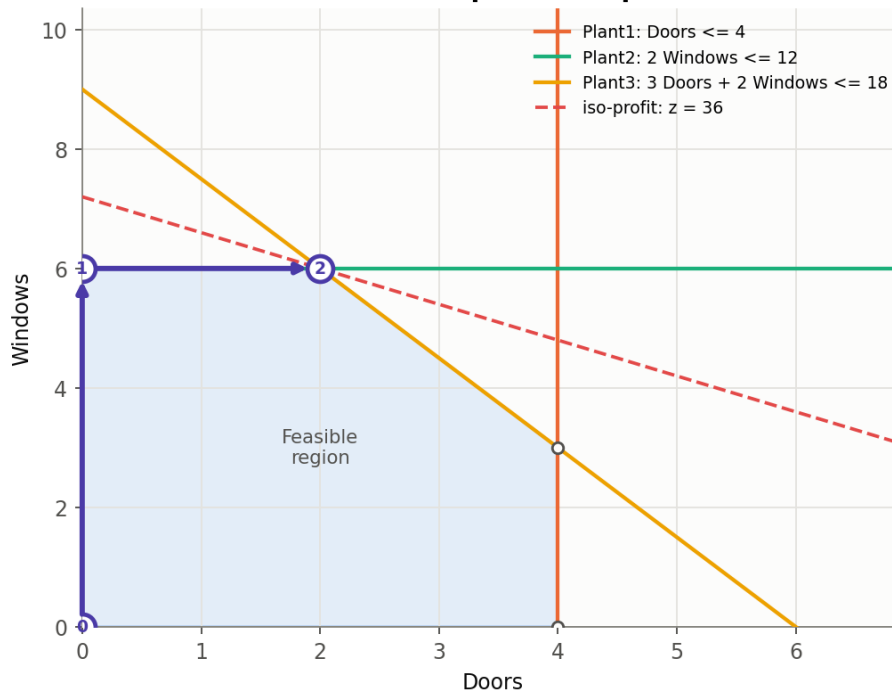
Graphical view of the primal

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_j - c_j$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	2	Windows	s2	0	yes	no	(0, 0)
1	primal	2	Doors	s3	30	yes	no	(0, 6)
2	primal	2			36	yes	yes	(2, 6)

Path of the primal simplex



Primal simplex - phase 2, tableau 0
enter Windows, leave s2 (min ratio 6)

	Doors	Windows	s1	s2	s3	RHS
z	-3	-5	0	0	0	0
s1	1	0	1	0	0	4
s2	0	2	0	1	0	12
s3	3	2	0	0	1	18

Primal simplex - phase 2, tableau 1
enter Doors, leave s3 (min ratio 2)

	Doors	Windows	s1	s2	s3	RHS
z	-3	0	0	2.5	0	30
s1	1	0	1	0	0	4
Windows	0	1	0	0.5	0	6
s3	3	0	0	-1	1	6

Primal simplex - phase 2, tableau 2
optimal tableau

	Doors	Windows	s1	s2	s3	RHS
z	0	0	0	1.5	1	36
s1	0	0	1	0.333	-0.333	2
Windows	0	1	0	0.5	0	6
Doors	1	0	0	-0.333	0.333	2